FIRESIDE




# SELF-HOSTED PERSONAL CLOUD STORAGE SYSTEM

~KURIAS JOJI

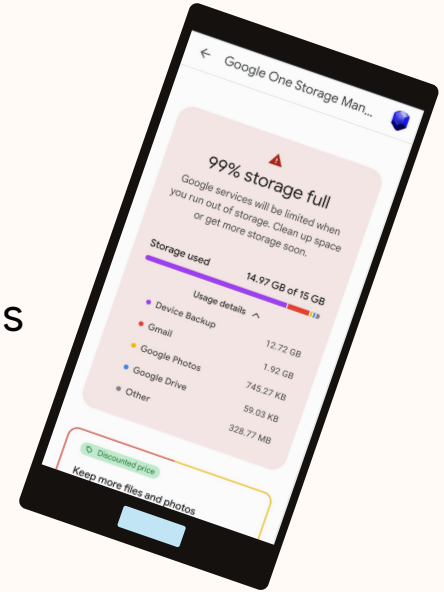
# The Problem

**Scattered Files**: Personal data is scattered across multiple devices.

**Third-Party Dependence**: Achieving centralized access currently requires relying on commercial cloud providers.

**Zero Control**: Users have no control over physical data locations, platform policies or how data is stored.

**The Cost Trap**: Arbitrary storage tiers force users into endless recurring subscriptions.



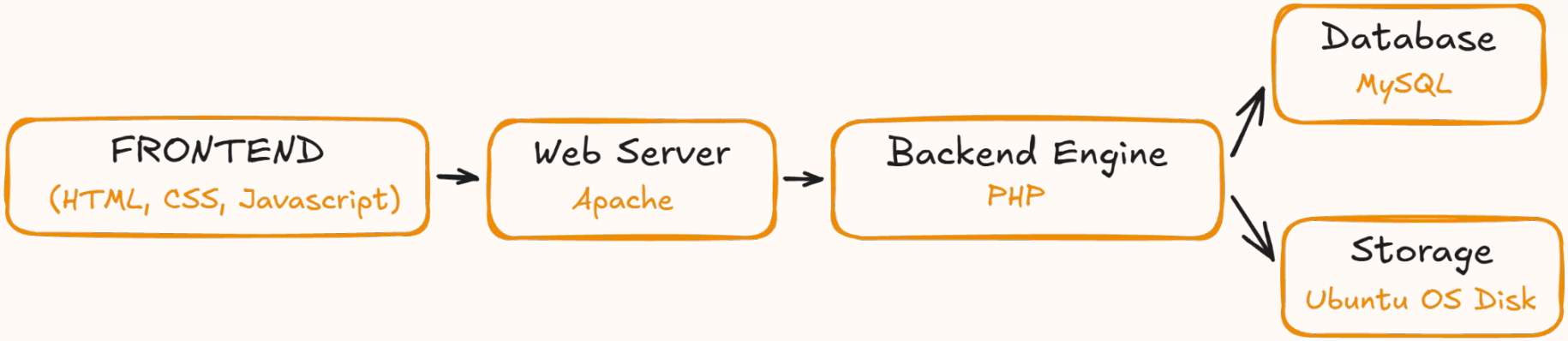
# Why a self-hosted cloud?

**Data Sovereignty**: Full control over where and how files are stored.
**Tailored Infrastructure**: Freedom to build the system to suit personal requirements.
**Hardware Sustainability**: Repurposing idle tech into a functional server.
**Zero Subscription Fees**: Eliminating monthly subscription costs by self-hosting.

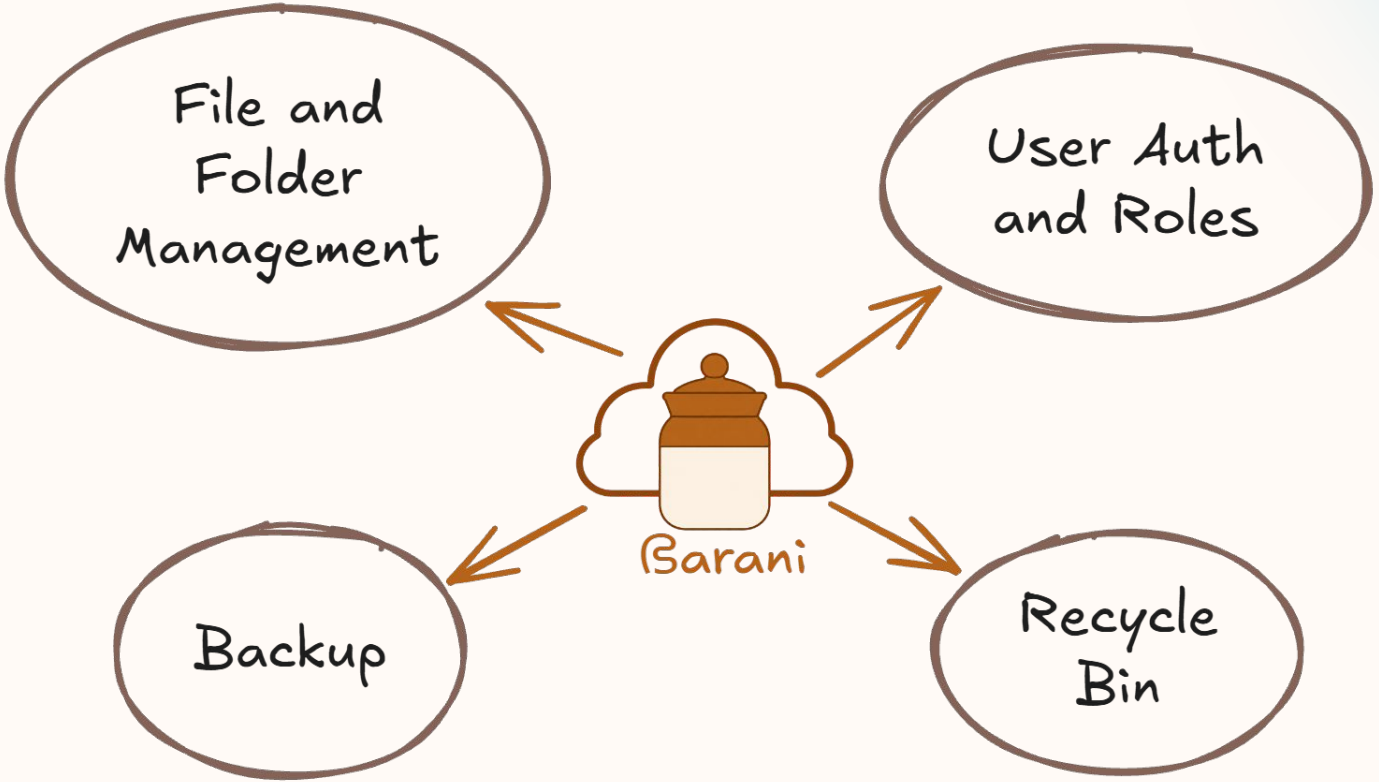


# Garani - A Self-Hosted Personal Cloud Storage System

A self-hosted personal cloud storage system that provides authenticated web-based access for securely storing, accessing, and backing up files on a user-controlled storage server.



# 4 Main Modules



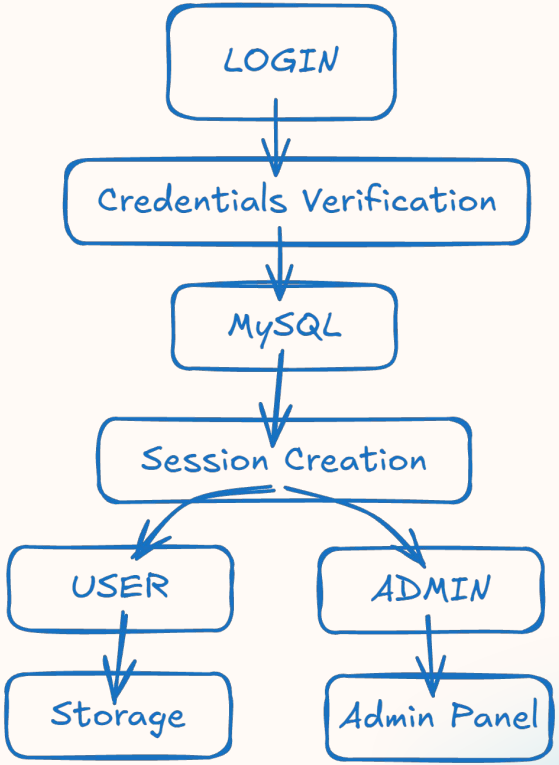
# User Authentication & Roles

**Secure Authentication**: Passwords are securely hashed and verified, with PHP sessions used for login.

**Role-Based Access Control**:

- **Admin**: Manages users and the system.
- **Standard User**: Accesses only their own storage.

**Session & Storage Security**: Sessions identify users, while path validation and storage isolation prevent unauthorized file access.



# File & Folder Management

| FEATURE | DESCRIPTION |
| --- | --- |
| Dynamic Navigation | Folders, Breadcrumbs |
| Filesystem Storage | Individual storage per User accessed via "FileSystemService" |
| File Operations | Folder Creation, File - upload, download, rename & delete |
| Deletion Workflow | Deleted folders or files are moved to Recycle Bin |



# Recycle Bin

**Soft Delete**: Deleted files are moved to a dedicated Recycle Bin instead of being permanently erased.

**Data Recovery**: Deleted files can be restored to their original location using stored file metadata.

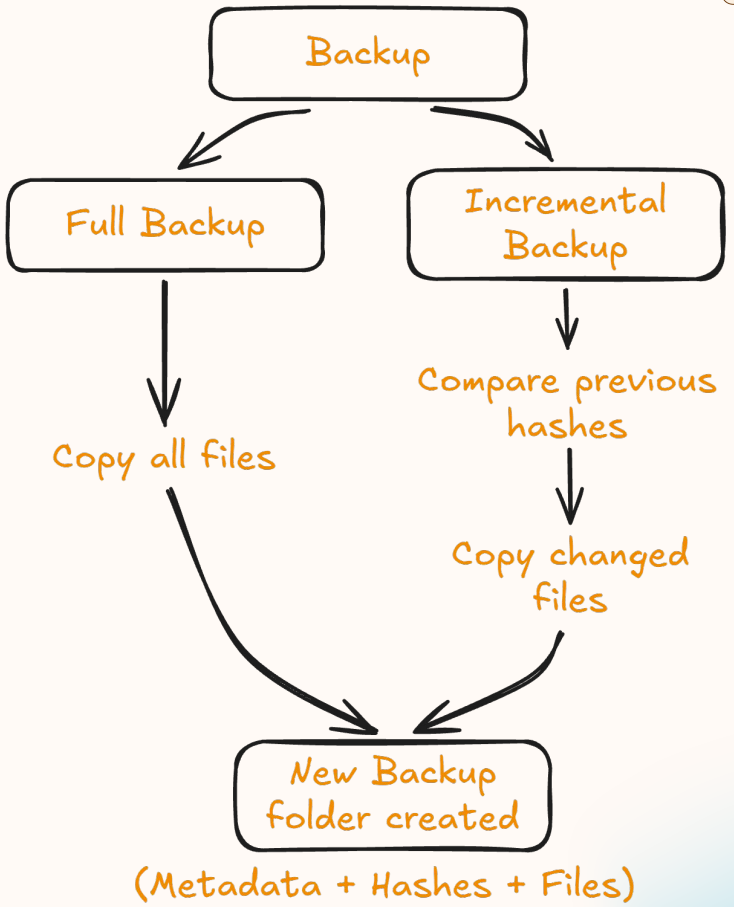
**Storage Reclamation**: Users can permanently delete items from here.

# Backups

**Full & Incremental Backups**: Creates complete backups or backs up only changed files.

**Integrity Verification**: Uses SHA-256 hashes to verify file integrity.

**Backup Metadata**: Stores backup details and file hashes in manifest.json and hashes.json.



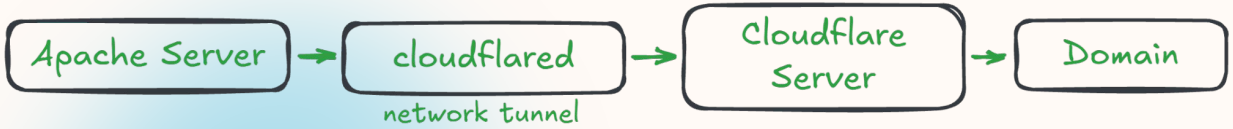
# External Client Access

## Initial Access Architecture

- **DHCP Reservation**: Assigned the storage server a fixed private IP.
- **Port Forwarding**: Initially planned to expose the server through the router for WAN access by pointing a domain to the router's public ip address.

## Pivot: Cloudflare Tunnel

- Moved away from direct port forwarding to avoid exposing router ports and to work around ISP CGNAT and dynamic IP limitations.
- Cloudflare Tunnel provides a secure connection between the server and the internet.




- Benefits: TLS encryption, and no open inbound ports.

# Basic Challenges

**Server Availability**: Maintaining 24/7 uptime and keeping hardware stable.

**Network & Remote Access**: Configuring secure HTTPS access and handling changing IPs/network issues.

**Data Security**: Protecting user files, authentication, and access permissions.

**Backup Restoration**: Preventing data loss and ensuring backups can actually be restored.

**Storage Management**: Managing disk space, user quotas, and storage growth.

# Long Term Goals



## File Search

Fast search across user storage



## File & Folder Sharing

Shareable links with access control



## Mobile Photo Sync

Automatic photo backup from mobile devices



# Thank You